Image by Number

Introduction

Computers store drawings, photographs and other pictures using only numbers. The following activity demonstrates how they can do this.

Computer screens are divided into a grid of small dots called pixels (**pic**ture **el**ements). In a black-and-white picture, each pixel is either black or white. In this activity, students explore how images can be represented using only 0s and 1s. They will also learn a method called Run-Length Encoding, which helps compress image data by reducing repetition.



Activity	Time	Description
Launch and Exploration	10 min	Introduce pixels and binary image representation. Demonstrate how a picture can be converted into 0s and 1s.
Discussions & Conclusion	25 min	Students work alongside the worksheet while the teacher guides the discussion. Student Worksheet: https://tinyurl.com/imagebynumberworksheet Students solve image encoding and decoding questions. Introduce Run-Length Encoding and practice shortening image representations. Discuss the worksheet. Discuss how computers store and compress images.

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CT Components

Algorithmic Thinking
Step-by-step procedures to encode and decode images. The process of converting a sequence of pixels to numbers and back to images.

Pattern Recognition
Identify repeated pixel patterns and sequences.

Decomposition
Images are broken into rows and individual pixels. Decompose an image into horizontal lines, then decompose each line into alternating encodings of white and black pixels.

Logic and Abstraction
Images are simplified to 0s and 1s. Encoding sequences of numbers is equivalent to a picture representation. Converting visual information into data that a computer can store. Understanding alternating colour groups.

Optimisation
Run-Length Encoding reduces repeated information to save space.

Encoding and Decoding
Translating an image into binary representation and back to an image.

Spatial Reasoning
Visualising the image from the numerical pattern.

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Activity: Image By Number

Every image we see on a screen, whether it is a picture, a video, or text, is displayed using pixels. Screens are divided into a grid of small dots called pixels (**pic**ture **el**ements).



In a black-and-white image, each pixel can be either black or white, so all the computer would need to store is which dots are black and which are white. For example, if we wanted to display the letter C, we would first divide it into squares. If we zoom in further and further on the letter, we can see a grid of pixels similar to these:

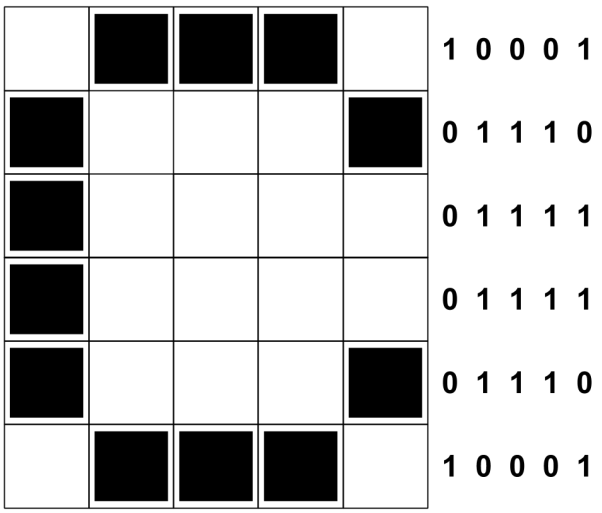


We can represent this image using the numbers 0 and 1.
A black-and-white image can be represented using only two values:
1 → White pixel
0 → Black pixel

If a 1 indicates a white square and a 0 indicates a black square, then we can represent our letter C on a 5x6 pixel grid, like this:
10001, 01110, 01111, 01111, 01110, 10001

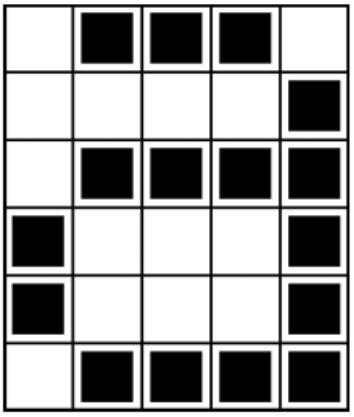
If we take these numbers and draw the image they represent, we get the letter C:

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The first line consists of one white pixel, then three black pixels, then one white pixel. Thus, the first line is represented as 1 0 0 0 1. The second line consists of one black pixel, then three white pixels, then one black pixel. Thus, the second line is represented as 0 1 1 1 0. Similarly, we can write the next four lines as 0 1 1 1 1, 0 1 1 1 1, 0 1 1 1 0 and 1 0 0 0 1.

Similarly, the letter “a” has been magnified to show the pixels.



Q. What will be the correct representation of image ‘ a ’ on a 5x6 pixel grid using 0s and 1s? 1 indicates a white square and a 0 indicates a black square?

a) 10111, 10111, 10001, 1010, 10101, 10001
b) 10001, 01110, 01111, 01111, 01110, 10001
c) 11110, 11110, 11000, 10110, 10110, 00111
d) 10001,11110, 10000, 01110, 01110, 10000

Answer b) 10001, 01110, 01111, 01111, 01110, 10001

Compare the visual pixel arrangement with the binary representation row by row.

 10001

CT Competency

Pattern Recognition: Identifying which binary pattern matches the image.
Abstraction: Understanding that numbers can represent pictures.
Logic: Comparing rows carefully to eliminate incorrect choices.

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Q. For the representation 00000, 11011, 11011, 11011, 11011, 11011, which of the following will be the alphabet in the image?

a) L
b) D
c) E
d) T

Answer d) T

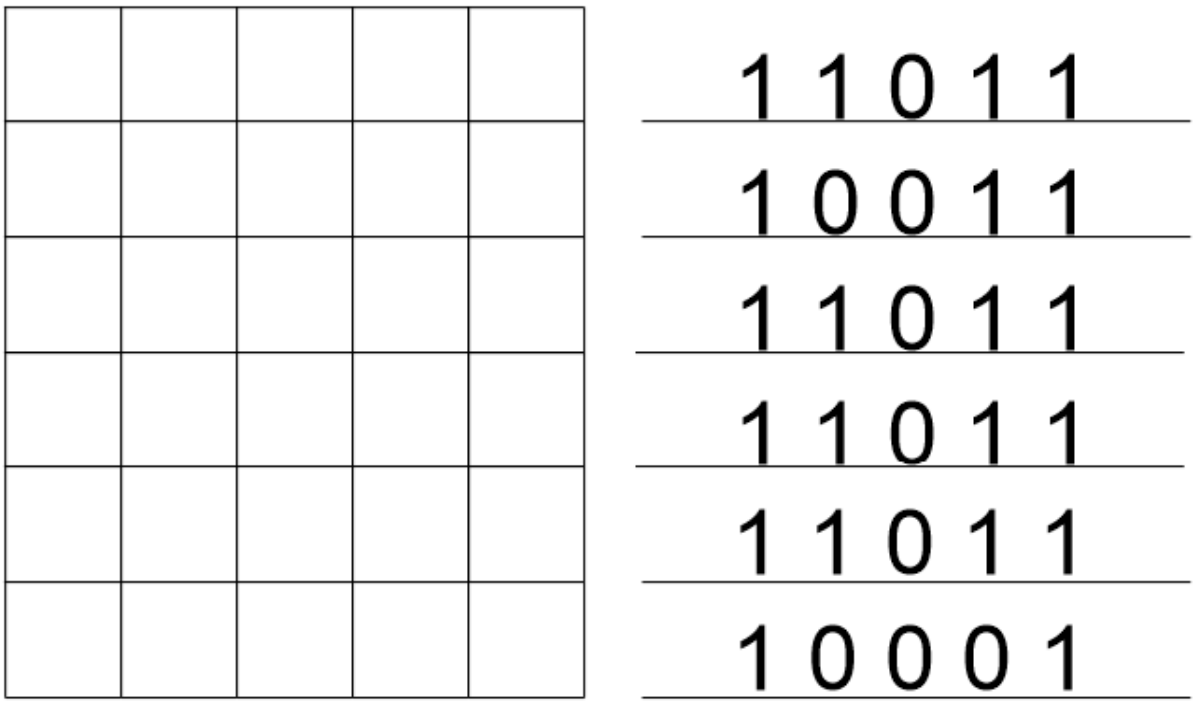
The first row forms a complete black line, and the middle column forms the vertical part of the letter T.

CT Competency

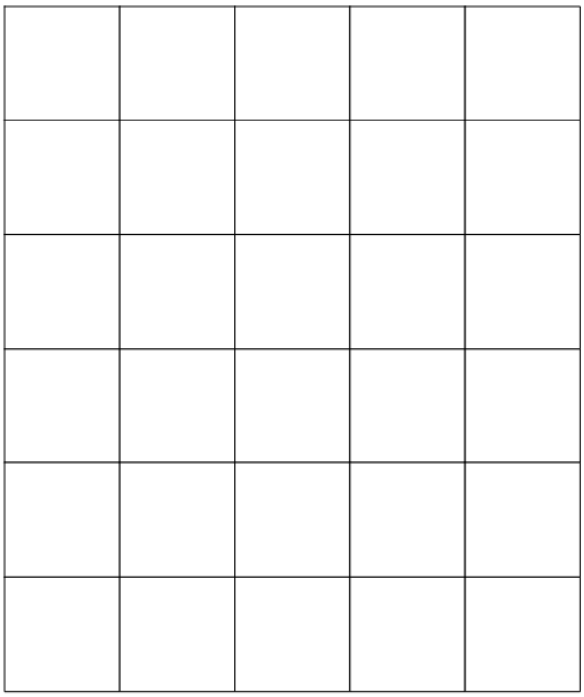
Decoding: Translating binary representation back into an image.
Spatial Reasoning: Visualising the image from the numerical pattern.
Algorithmic Thinking: Following a systematic row-by-row process.

The picture above shows us how an image can be represented by numbers.

Draw the Image using the given representation :

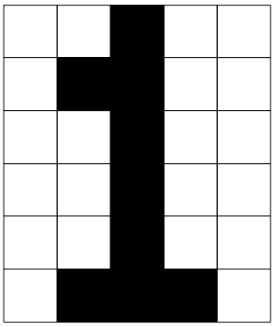


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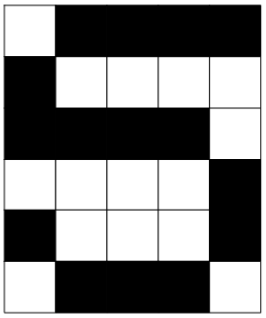


10000
01111
00001
11110
01110
10001

Answer:



11011
10011
11011
11011
11011
10001



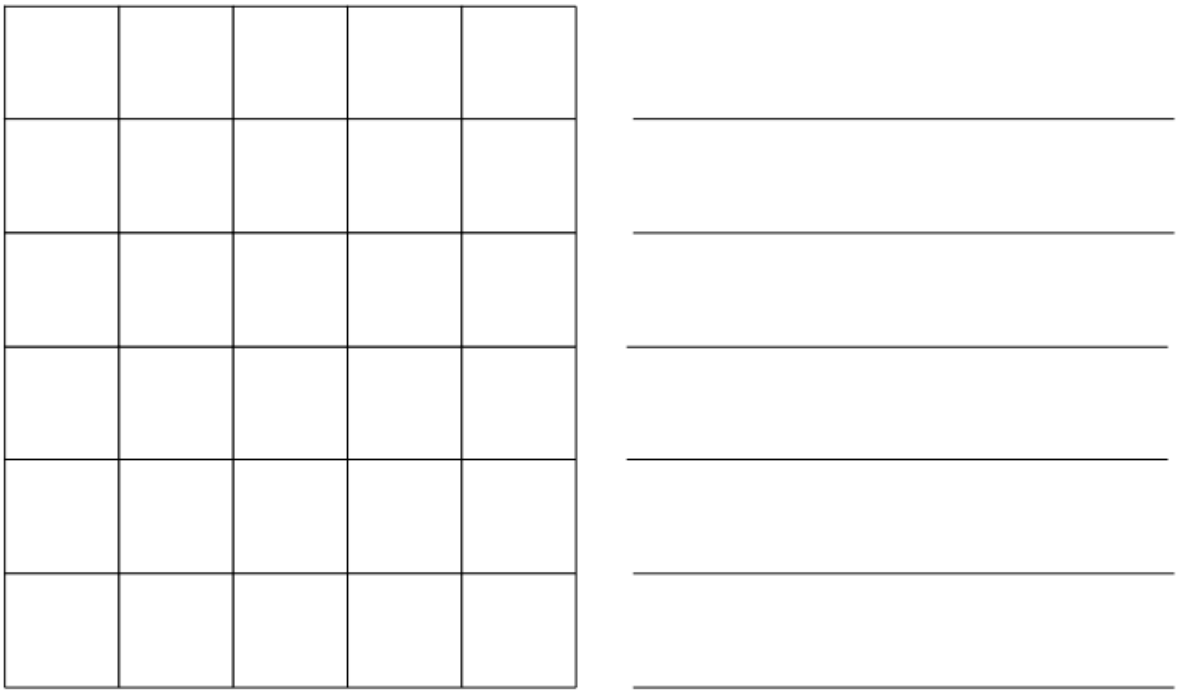
10000
01111
00001
11110
01110
10001

CT Competency

Algorithmic Thinking: Step-by-step procedures to decode images. The process of converting a sequence of numbers to images.
Spatial Reasoning: Visualising the image from the numerical pattern.
Logic and Abstraction: Images are simplified to 0s and 1s. Encoding sequences of numbers is equivalent to a picture representation.

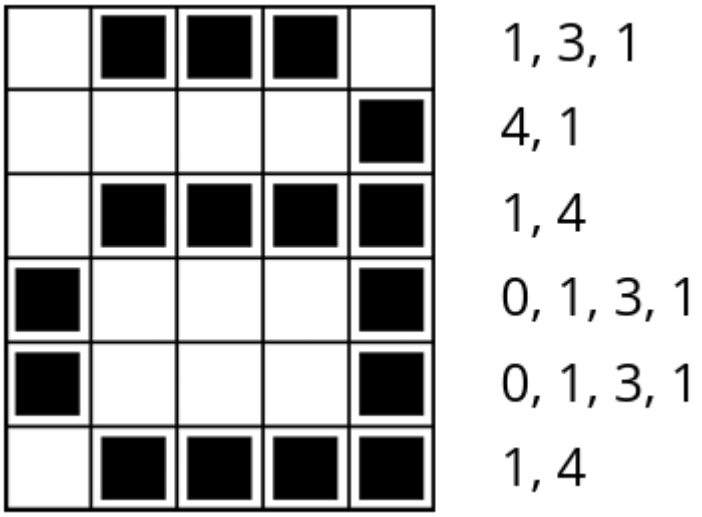
Create the first letter of your name using a 5×6 pixel grid. And write its binary representation.

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Now, instead of representing every pixel with 0s and 1s. We can shorten the representation by alternatively writing how many white and black pixels are needed in each line.

We will start by mentioning the number of white pixels, then the number of black pixels, then the number of white pixels again, and so on.



The first number always relates to the number of white pixels. If the first pixel is black, the line will begin with a zero. As shown in Line 4, the image starts with a black pixel, so we need to write 0 for the number of white pixels, then write the number of black pixels.

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This concept of shortening the representation, known as Run Length Encoding, encodes each continuous group of white and black pixels (or 1's or 0's) encountered in a left-to-right scan of a row by its length.

Instead of writing **10001**, we can write **131**: 1 white pixel, 3 black pixels, and 1 white pixel.

Q. Convert the following row into Run-Length Encoding.

11110000 : _____

Answer 11110000: 4 white pixels and 4 black pixels : 44

CT Competency
Pattern Recognition: Detecting repeated sequences.
Optimisation: Compressing repeated data.
Algorithmic Thinking: Following encoding rules step by step.

Q. Decode the following Run-Length

221 : _____

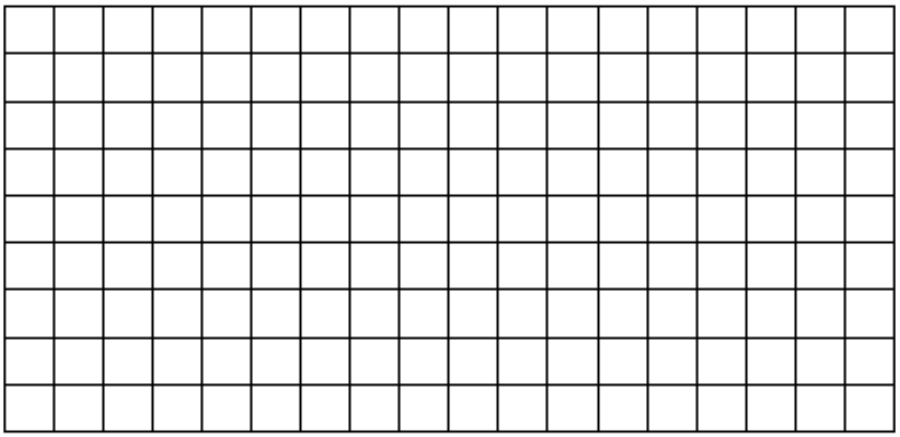
Answer 221 : 2 white pixels, 2 black pixels, 1 white pixel : 11001

CT Competency
Decoding: Reconstructing information from encoded data.
Logic: Understanding alternating colour groups.

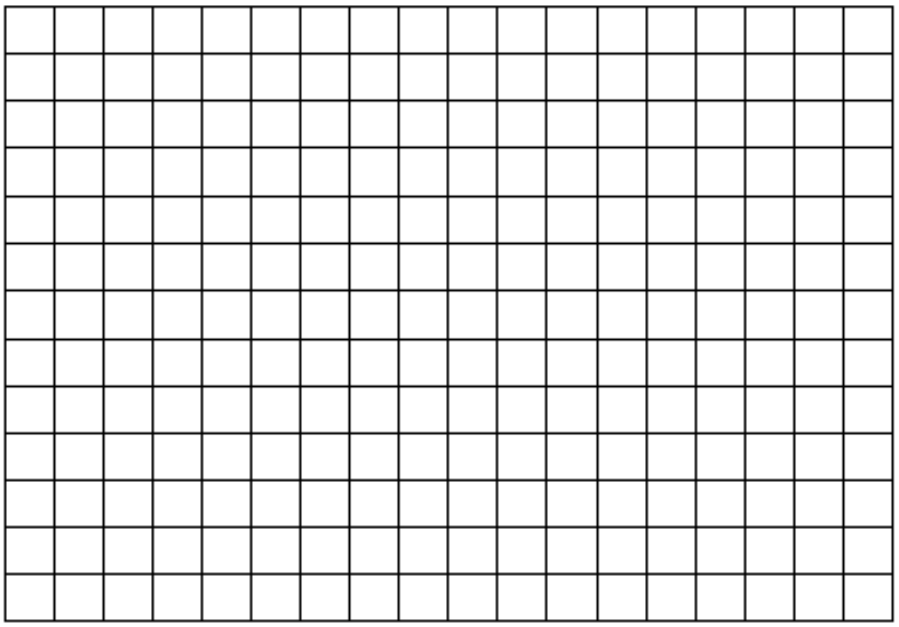
This process was used for black-and-white scanned images on fax machines, which use this approach to compression. Fax machines were used to transmit an exact copy of a paper document to another fax machine.

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Draw the following:

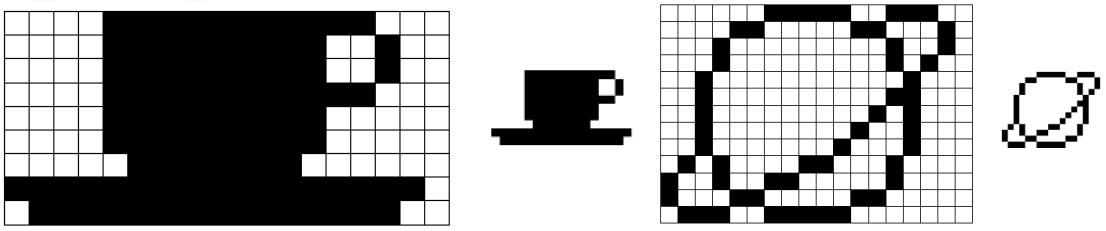


4, 11
4, 9, 2, 1
4, 9, 2, 1
4, 11
4, 9
4, 9
5, 7
0, 17
1, 15



6, 5, 2, 3
4, 2, 5, 2, 3, 1
3, 1, 9, 1, 2, 1
3, 1, 9, 1, 1, 1
2, 1, 11, 1
2, 1, 10, 2
2, 1, 9, 1, 1, 1
2, 1, 8, 1, 2, 1
2, 1, 7, 1, 3, 1
1, 1, 1, 1, 4, 2, 3, 1
0, 1, 2, 1, 2, 2, 5, 1
0, 1, 3, 2, 5, 2
1, 3, 2, 5

Answer:



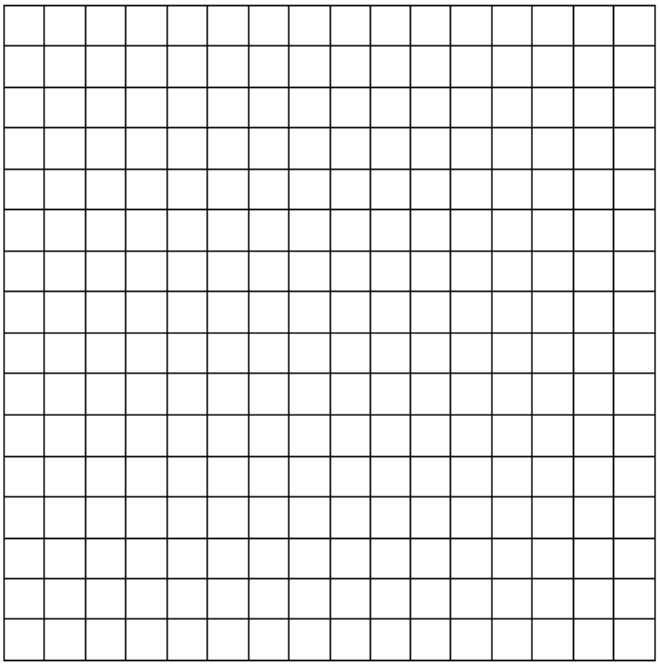
CT Competency

Algorithmic Thinking: Step-by-step procedures to decode images.
Spatial Reasoning: Visualising the image from the numerical pattern.
Decoding: Reconstructing information from encoded data.
Logic: Understanding alternating colour groups.

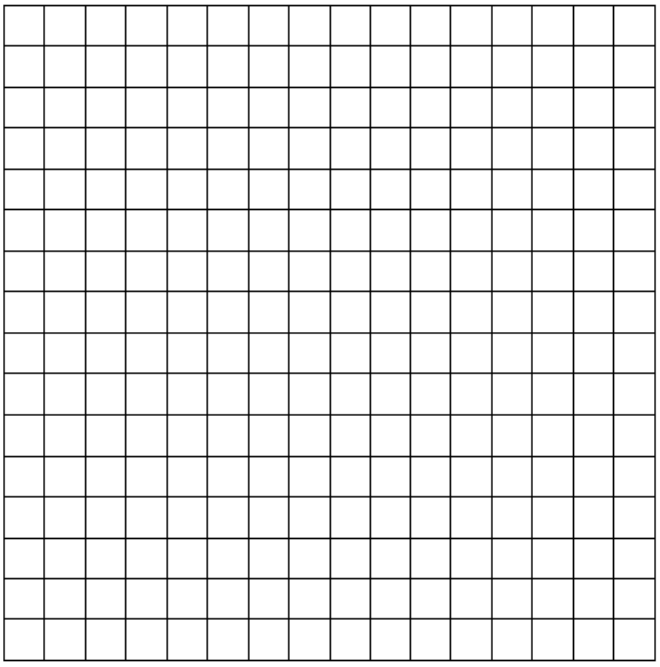
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Pattern Recognition: Detecting repeated sequences.

Draw the image you want to send to your friend, in the provided grid, and write its Run-Length Encoding. Exchange Run-Length Encoded messages with your friend.



Draw the image your friend sent, in the provided grid, by decoding its Run-Length Encoding.



Computers represent images using numbers. By converting images into binary patterns and compressing them using Run-Length Encoding, we have discovered how digital systems store and communicate visual information efficiently.

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